PARCEL SORTING AND DISPATCHING SYSTEM

ABSTRACT

The Parcel Sorting and Dispatching System is a simulation-based application developed using the C programming language to demonstrate the practical implementation of data structures in parcel management and logistics operations. The main objective of this project is to simulate the process of receiving, sorting, tracking, and dispatching parcels efficiently using appropriate data structures.

The system simulates the arrival of parcels, each containing information such as a unique parcel ID, destination city, priority, size, and arrival time. Incoming parcels are managed using a Queue based on the First-In-First-Out (FIFO) principle. A Binary Search Tree (BST) is used to organize parcels according to their destination cities, while a Hash Table enables efficient parcel tracking and status updates. A Circular Linked List manages the rotation of dispatch terminals, ensuring that different destinations receive their turn for parcel dispatch. Parcels that are incorrectly routed are temporarily stored in a Stack and reprocessed for dispatch.

The application operates through a time-based simulation in which parcel arrivals, sorting operations, terminal rotations, and dispatch activities are performed at regular intervals. It also generates reports containing details such as the total number of parcels processed, successfully dispatched parcels, returned parcels, and processing delays.

This project demonstrates how different data structures can work together to manage parcel flow, organize logistics operations, and improve the efficiency of a simulated parcel sorting and dispatching system.

DATA STRUCTURES USED

Data Structure	Purpose in the Project
Queue	Manages incoming parcels in First-In-First-Out (FIFO) order.
Stack	Stores misrouted or returned parcels temporarily for reprocessing using Last-In-First-Out (LIFO) order.
Binary Search Tree (BST)	Sorts and organizes parcels according to their destination cities for efficient retrieval and dispatch.
Hash Table	Enables fast searching, tracking, and updating of parcel information using unique parcel IDs.
Circular Linked List	Manages the rotation of dispatch terminals, allowing different destination cities to receive parcels in a cyclic order.